Sunacef Powder for Injection

[Ingredient]
Each vial contains：
Cefuroxime (as Sodium)…………750 mg (potency, for pack of 750 mg)
Cefuroxime (as Sodium)…………1.5 g (potency, for pack of 1.5 g)

[Description]
It is a white to pale yellow powderin a seal vial. The color of the vial is colorless.
After reconstitution, constituted solutions may range in color from light yellow to amber. The solutions darken, depending on storage conditions, without affecting their potency.

[Pharmacodynamics]
Microbiology
Cefuroxime has in vitro activity against a wide range of gram-positive and gram-negative organisms, and it is highly stable in the presence of beta-lactamases of certain gram-negative bacteria. The bactericidal action of cefuroxime results from inhibition of cell-wall synthesis.

Cefuroxime is usually active against the following organisms in vitro.

Aerobes, Gram-positive: Staphylococcus aureus, Staphylococcus epidermidis, Streptococcus pneumoniae, and Streptococcus pyogenes (and other streptococci). NOTE: Most strains of enterococci, e.g., Enterococcus faecalis (formerly Streptococcus faecalis), are resistant to cefuroxime. Methicillin-resistant staphylococci and Listeria monocytogenes are resistant to cefuroxime.

Aerobes, Gram-negative: Citrobacter spp., Enterobacter spp., Escherichia coli, Haemophilus influenzae (including ampicillin-resistant strains), Haemophilus parainfluenzae, Klebsiella spp. (including Klebsiella pneumoniae), Moraxella (Branhamella) catarrhalis (including ampicillin- and cephalothin-resistant strains), Morganella morganii (formerly Proteus morganii), Neisseria gonorrhoeae (including penicillinase- and non-penicillinase-producing strains), Neisseria meningitidis, Proteus mirabilis, Providencia rettgeri (formerly Proteus rettgeri), Salmonella spp., and Shigella spp.

NOTE: Some strains of Morganella morganii, Enterobacter cloacae, and Citrobacter spp. have been shown by in vitro tests to be resistant to cefuroxime and other cephalosporins. Pseudomonas and Campylobacter spp., Legionella spp., Acinetobacter calcoaceticus, and most strains of Serratia spp. and Proteus vulgaris are resistant to most first- and second-generation cephalosporins.

Anaerobes: Gram-positive and gram-negative cocci (including Peptococcus and Peptostreptococcus spp.), gram-positive bacilli (including Clostridium spp.), and gram-negative bacilli (including Bacteroides and Fusobacterium spp.).

NOTE: Clostridium difficile and most strains of Bacteroides fragilis are resistant to cefuroxime.

[**Pharmacokinetics]**
Therapeutic serum concentrations of approximately 2 mcg/mL or more are maintained for 5.3 hours and 8 hours or more, respectively. There is no evidence of accumulation of cefuroxime in the serum following IV administration of 1.5-g doses every 8 hours to normal volunteers. The serum half-life after either IM or IV injections is approximately 80 minutes.

Approximately 89% of a dose of cefuroxime is excreted by the kidneys over an 8-hour period, resulting in high urinary concentrations.

Following the IM administration of a 750-mg single dose, urinary concentrations averaged 1,300 mcg/mL during the first 8 hours. Intravenous doses of 750 mg and 1.5 g produced urinary levels averaging 1,150 and 2,500 mcg/mL, respectively, during the first 8-hour period.

The concomitant oral administration of probenecid with cefuroxime slows tubular secretion, decreases renal clearance by approximately 40%, increases the peak serum level by approximately 30%, and increases the serum half-life by approximately 30%. Cefuroxime is detectable in therapeutic concentrations in pleural fluid, joint fluid, bile, sputum, bone, and aqueous humor.

Cefuroxime is detectable in therapeutic concentrations in cerebrospinal fluid (CSF) of adults and pediatric patients with meningitis.

[Incompatibilities]
Cefuroxime is compatible with most commonly used intravenous fluids and electrolyte solutions.

The pH of 2.74% w/v sodium bicarbonate injection BP considerably affects the colour of solutions and therefore this solution is not recommended for the dilution of Cefuroxime for Injection. However, if required, for patients receiving sodium bicarbonate injection by infusion the Cefuroxime for Injection may be introduced into the tube of the giving set.

Cefuroxime for Injection should not be mixed in the syringe with aminoglycoside antibiotics.

[Indications]
Cefuroxime for Injection is a bactericidal cephalosporin antibiotic which is resistant to most ß-lactamases and is active against a wide range of Gram-positive and Gram-negative organisms. It is indicated for the treatment of infections before the infecting organism has been identified or when caused by sensitive bacteria. Indications include:- Respiratory tract infections for example, acute and chronic bronchitis, infected bronchiectasis, bacterial pneumonia, lung abscess and post-operative chest infections. Ear, nose and throat infections for example, sinusitis, tonsillitis, pharyngitis and otitis media. Urinary tract infections for example, acute and chronic pyelonephritis, cystitis and asymptomatic bacteriuria. Soft-tissue infections for example, cellulitis, erysipelas and wound infections. Bone and joint infections for example, osteomyelitis and septic arthritis. Obstetric and gynaecological infections, pelvic inflammatory diseases. Gonorrhoea particularly when penicillin is unsuitable. Other infections including septicaemia, meningitis and peritonitis. Prophylaxis against infection in abdominal, pelvic, orthopaedic, cardiac, pulmonary, oesophageal and vascular surgery where there is increased risk from infection. Usually Cefuroxime for Injection will be effective alone, but when appropriate it may be used in combination with an aminoglycoside antibiotic, or in conjunction with metronidazole (orally or by suppository or injection), especially for prophylaxis in colonic or gynaecological surgery.

[Recommended Dose]
Usually cefuroxime is effective when administered alone, but when appropriate it may be used in combination with metronidazole or an aminoglycoside.

General Dosage

Adults: Many infections will respond to 750mg three times daily by intramuscular or intravenous injection. For more severe infections this dose should be increased to 1.5g three times daily intravenously. The frequency of dosage may be increased to six-hourly injections, intramuscular or intravenous, giving total daily doses of 3g to 6g.

Infants and children: Doses of 30 to 100mg/kg/day given in three or four divided doses. A dose of 60mg/kg/day will be appropriate for most infections.

Neonates: Doses of 30 to 100mg/kg/day given in two or three divided doses. In the first weeks of life the serum half-life of cefuroxime can be three to five times that in adults.

Gonorrhoea

1.5g should be given as a single dose or as two 750mg injections into different sites, eg, each buttock.

Meningitis

Cefuroxime therapy is suitable for sole therapy of bacterial meningitis due to sensitive strains.

Infants and children: 200 to 240mg/kg/day intravenously in three or four divided doses. This dosage may be reduced to 100mg/kg/day after three days or when clinical improvement occurs.

Neonates: The initial dosage should be 100mg/kg/day intravenously. This dosage may be reduced to 50mg/kg/day after three days or when clinical improvement occurs.

Adults: 3g intravenously every eight hours. No data is currently available to recommend a dose for intrathecal administration.

Prophylaxis

The usual dose is 1.5g intravenously with induction of anaesthesia. For orthopaedic, pelvic and abdominal operations this may be followed with two 750mg doses 8 and 16 hours later. For vascular, cardiac, oesophageal and pulmonary operations this may be supplemented with 750mg intramuscularly three times a day for a further 24 to 48 hours.

In total joint replacement, 1.5g cefuroxime powder may be mixed dry with each pack of methyl methacrylate cement polymer before adding the liquid monomer.

Dosage in Impaired Renal Function

As cefuroxime is excreted by the kidneys, the dosage should be reduced to allow for slower excretion in patients with impaired renal function, once creatinine clearance falls below 20ml/min, as follows:

Marked impairment (creatinine clearance 10 to 20ml/min)	750mg twice daily
Severe impairment (creatinine clearance of less than 10ml/min)*	750mg once daily
Continuous peritoneal dialysis	750mg twice daily
Renal failure on continuous arteriovenous haemodialysis or high-flux haemofiltration in intensive therapy units	750mg twice daily
Low-flux haemofiltration	As for impaired renal function

*For patients on haemodialysis, a further 750mg should be given at the end of each dialysis session.

[Mode of Administration]

Intramuscular
Add 3ml water for injection to Cefuroxime 750mg. Shake gently to produce an opaque suspension.

Intravenous
Dissolve Cefuroxime in water for injection using at least 6ml for 750mg or 15ml for 1.5g. For short intravenous infusion (e.g. up to 30 minutes), 1.5g may be dissolved in 50ml water for injection. These solutions may be given directly into the vein or introduced into the tubing of the giving set if the patient is receiving parenteral fluids.

[Contraindications]
Cefuroxime for Injection is contraindicated in patients with known allergy to the cephalosporin group of antibiotics.

[Warnings and Precautions]
Cephalosporin antibiotics may, in general, be given safely to patients who are hypersensitive to penicillins, although cross-reactions have been reported. Special care is indicated in patients who have experienced an anaphylactic reaction to penicillin.

Cephalosporin antibiotics at high dosage should be given with caution to patients receiving potent diuretics or aminoglycosides, as these combinations are suspected of adversely affecting renal function. Clinical experience has shown that this is not likely to be a problem at the recommended dose levels.

Serious and occasionally fatal hypersensitivity reactions (including anaphylactoid and severe cutaneous adverse reactions) have been reported in patients receiving therapy with bata-lactams. Before initiating therapy with Sunacef, careful inquiry should be made concerning previous hypersensitivity reactions to penicillins, cephalosporins, carbapenems or other beta-lactam agents. If an allergic reaction occurs, Sunacef must be discontinued immediately and appropriate alternative instituted.

[Effects on ability to drive and use machines]
No studies on the effects of cefuroxime on the ability to drive and use machines have been performed. However, based on known adverse reactions, cefuroxime is unlikely to have an effect on the ability to drive and use machines.

[Interaction with other medicinal products]
Concurrent administration of probenecid prolongs the excretion of cefuroxime and produces an elevated peak serum level.

Concurrent administration of potent diuretics, aminoglycosides may adversely affect renal function.
<u>Interference with Laboratory Tests</u>

Slight interference may occur with the copper reduction methods (Fehling's, Benedict's) but this should not lead to false-positive results. Cefuroxime does not interfere with the enzyme based tests for glycosuria, or with the alkaline picrate method for creatinine. It is recommended that either the hexokinase or glucose oxidase methods are used for determination of blood/plasma glucose levels.

[Pregnancy and Lactation]
<u>Pregnancy</u>
Teratogenic Effects: Pregnancy Category B. Reproduction studies have been performed in mice and have revealed no evidence of impaired fertility or harm to the fetus due to cefuroxime. There are, however, no adequate and well-controlled studies in pregnant women. Because animal reproduction studies are not always predictive of human response, this drug should be used during pregnancy only if clearly needed.

<u>Nursing Mothers</u>
Since cefuroxime is excreted in human milk, caution should be exercised when Cefuroxime for Injection is administered to a nursing woman.

[Side Effects]
Hypersensitivity reactions: Including skin rashes (maculopapular and urticarial), interstitial nephritis, drug fever and, very rarely, anaphylaxis. As with any antibiotic, prolonged use may lead to overgrowth of non-susceptible organisms, eg, Candida.

As with other cephalosporins, there have been rare reports of erythema multiforme, Stevens-Johnson syndrome and toxic epidermal necrolysis.

Gastro-intestinal disturbance: Including, very rarely, pseudomembranous colitis, which has been reported with most broad spectrum antibiotics.

Haematological: A decrease in haemoglobin concentration, eosinophilia, leucopenia and neutropenia have been observed. Positive Coombs' tests have been reported. As with other cephalosporins, thrombocytopenia has been reported rarely.

Hepatic: Transient rises in liver enzymes or serum bilirubin have been observed, particularly in patients with pre-existing liver disease, but there is no evidence of hepatic involvement.

Renal: There may be some variation in the results of biochemical tests of renal function, but these results do not appear to be of clinical significance.

Other: Transient pain may be experienced at the site of intramuscular injection. Occasionally thrombophlebitis may occur at the site of intravenous injection. A burning sensation may be observed after intravenous injection. Mild to moderate hearing loss has been reported in some children treated for meningitis. Dizziness and headache has been reported in patients receiving cefuroxime.

[Toxicity and treatment of overdose]
Overdosage of cephalosporins can cause cerebral irritation leading to convulsions. Serum levels of cefuroxime can be reduced by hemodialysis and peritoneal dialysis.

[Package]
750 mg × 50's vials
1.5 g × 50's vials

[Storage]
Store at temperature below 30℃, protect from light. Reconstituted solutions may be stored for up to 4 hours at room temperature (25-30℃) or 48 hours under refrigeration (5-10℃). Unused solutions should be discarded after the time periods mentioned above. Discard any unused portion after opening.

[Shelf-life]

Pack of 750 mg:	Three years
Pack of 1.5 g:	Two years

Registration No.:	Pack of 750 mg:	MAL20152506AZ
	Pack of 1.5 g:	MAL20152505AZ

Manufacturer:
TAIWAN BIOTECH CO., LTD. KUAN YIN PLANT
No. 1, Kwo Chieh 1st Road, Kuan Yin District, Taoyuan City, Taiwan, R.O.C.

Importer & Product Registration Holder:
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